

Exercise 9: D3 and Observable

Three notebook sections, three captions and a reflection. No separate document.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv, added to your notebook's files
Tools	Observable (observablehq.com), free account. D3 and Inputs are available in every notebook.
Resources	Observable getting-started guide; Notebooks 2.0 system guide; D3 API reference (d3js.org/api). Learning Tableau 2025 doesn't cover D3.
Walkthrough	Module 9 of the MVU Tableau Walkthrough has every code cell, ready to copy, with checkpoints.
Submit	A link to your Observable notebook that your instructor can open.

Before you start

In Tableau, dropping a measure on a shelf chose the scale, drew the axis and started it at zero. In D3 you write each of those decisions yourself. This exercise builds the same graduation-rate chart you know from earlier modules, in code, so you can see which decisions Tableau was making for you.

Two versions of Observable. The walkthrough code uses the current editor (Notebooks 2.0): `const x = ...` and `view(...)`. In the older editor, write `data = ...` without `const`, `viewof minGPA = Inputs.range(...)`, and end a code block with `return svg.node()` instead of `display(...)`.

Section 1: load, inspect, and transform

1. Create a notebook named DIV 6380 Exercise 9 and add mvu_student_dim.csv to its files.
2. Load it with typed values, and show it as a table:

```
const data = FileAttachment("mvu_student_dim.csv").csv({typed: true});
```
3. Filter to the 2020 and 2021 cohorts, then use `d3.rollup` to compute the mean of `is_graduated` by college.
4. Convert the rollup to an array of `{college, rate, n}`, sorted by rate, and show it with `Inputs.table`.

Checkpoint: College of Health Sciences graduation rate, 2020–21 cohorts?

Section 2: a bar chart you build by hand

1. Define width, height and margins.
2. Build `x` with `d3.scaleBand` (one band per college) and `y` with `d3.scaleLinear`, domain `[0, 1]`.
3. Bind rates to `rect` elements with `.data(rates).join("rect")`; draw both axes; label each bar with its rate and `n`.
4. Try the `y` domain as `d3.extent` of the rates, note what happens, then put `[0, 1]` back.

Checkpoint: How many bars does the chart draw?

Section 3: inputs that drive the chart

1. Add a checkbox input for cohorts (2020, 2021, 2022; default 2020 and 2021) and a range input for minimum undergrad GPA (2 to 4, step 0.1).
2. Change the filter so it reads both inputs. Everything downstream updates on its own.
3. Tick 2022 and watch what happens. Decide what your notebook should do about it.

Checkpoint: With 2020–21 cohorts and minimum GPA 3.5, how many students are in the filtered data?

Captions and reflection

Use Markdown cells: a title and one-sentence description at the top, a heading before each section, and a caption after each.

Cell	What it must answer
Section 1 caption	What does typed: true change, and what would go wrong without it? Which Tableau step does the rollup replace?
Section 2 caption	What happened when you changed the y domain, and why did you keep the one you kept? What does the n label tell a reader that the bar can't?
Section 3 caption	What happens when a reader ticks 2022, and what did you do about it?
Reflection	Compare building this chart in D3 with building it in Tableau. Which decisions did Tableau make for you, which took longest here, and when would you choose each tool? About 250 words.

The reflection carries 25% on its own. “D3 is harder but more flexible” does not earn full marks. Name specific decisions (the zero baseline, the scale type, binding data to marks, reactivity) and say where each tool put them.

Before you submit: open your notebook link in a private browser window to check your instructor can see it.

Submit for Exercise 9

- A link to your Observable notebook, openable by your instructor
- Section 1: data loaded with typed: true, the 2020–21 filter, a rollup and a table
- Section 2: a D3 bar chart with band and linear scales, both axes, a zero baseline and n labels
- Section 3: cohort and GPA inputs driving the chart reactively
- A Markdown caption after each section, and the reflection at the end